

Notes: Antràs and Staiger (AER, 2012)
Offshoring and the Role of Trade Agreements

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1 Big picture

This paper asks how the rise of **offshoring of customized intermediate inputs** changes the logic of trade policy and the design of trade agreements. Traditional trade-agreement theory often studies final-good trade in which international prices are determined by market clearing. In that world, GATT/WTO principles such as market access, reciprocity, and nondiscrimination have a natural efficiency interpretation.

Antràs and Staiger argue that offshoring changes the problem. In their model, intermediate inputs are customized to a particular buyer, contracts are incomplete, and the price of the input is determined by **bilateral bargaining** after relationship-specific investment has already occurred. This lock-in feature creates an international hold-up problem and also changes the way governments can shift costs onto trading partners.

The core message is:

- Free trade in a customized intermediate input can lead to **too little input trade** because foreign suppliers anticipate ex post bargaining and underinvest.
- Trade policy can partly substitute for missing contractual safeguards by subsidizing input trade.
- Noncooperative trade policy is inefficient in two ways: it produces too little intermediate-input trade and distorts the home-market price of the final good.
- Efficient trade agreements in an offshoring world require **deep integration**: they must constrain policies beyond traditional market-access commitments.
- With political economy motives, the agreement must solve not only a terms-of-trade problem but also an additional political externality created by cross-border input bargaining.

Interpretation: The paper is not simply saying that offshoring increases trade in inputs. It says that offshoring changes the institutional problem of trade agreements, because bilateral bargaining over customized inputs breaks the market-clearing discipline that supports traditional WTO rules.

2 Incremental contribution of the paper

2.1 Conceptual contribution

The paper introduces a new reason why international trade agreements may need to look different when trade is organized through global value chains. Previous terms-of-trade theories supported shallow integration centered on tariff concessions and market-access commitments. This paper shows that once international input prices are bargained bilaterally rather than determined only by market clearing, market-access commitments are generally insufficient.

2.2 Theoretical contribution

The paper builds a tractable model with:

- a home country with final-good producers;
- a foreign country with intermediate-input suppliers;

- relationship-specific input investment;
- incomplete contracts;
- ex post Nash bargaining over the input price;
- trade-policy instruments on both intermediate and final goods.

The model delivers sharp propositions about free trade, efficient policy, Nash policy, and the scope of trade agreements.

2.3 Policy contribution

The paper explains why offshoring may push the world trading system away from simple tariff bargaining and toward deeper, more customized institutional arrangements. The result helps rationalize why preferential trade agreements, value-chain agreements, and behind-the-border rules have become more important as global production has fragmented.

Incremental message: In standard final-good trade models, the problem is mainly that countries restrict market access to improve their terms of trade. In this paper, the problem is broader: because input prices are bargained, many policies can affect the division of surplus and the level of relationship-specific investment. This makes deep integration more than an optional add-on; it becomes necessary for efficiency.

3 Data and motivating empirical pattern

The paper is primarily theoretical, but it provides one suggestive empirical fact. The authors examine tariff concessions for countries joining the WTO after 1995 and compare sectors with different degrees of input customization. Input customization is measured using Nunn’s input-contractibility measure: industries with a higher share of inputs not traded on organized exchanges are treated as more customized.

The empirical pattern is simple: WTO tariff concessions are larger in low-customization sectors and smaller, even negative relative to the mean, in high-customization sectors.

Interpretation: The evidence is not presented as a definitive test of the model. It is used to motivate the possibility that traditional WTO bargaining is less effective in sectors where inputs are relationship-specific and offshoring-like frictions are stronger.

4 Model environment

4.1 Countries and goods

The benchmark model has two small countries, Home H and Foreign F , plus a large rest of the world that fixes the world price of the final good. Preferences are quasi-linear:

$$U^j = c_0^j + u(c_1^j), \quad j \in \{H, F\}.$$

There are two main traded objects:

- a homogeneous final good 1;
- a customized intermediate input x used to produce the final good.

The final good is produced according to

$$y(x), \quad y'(x) > 0, \quad y''(x) < 0.$$

The marginal cost of producing the input in Foreign is normalized to one. Efficient input production requires

$$y'(x^E) = 1.$$

4.2 Incomplete contracts and relationship specificity

The key assumption is that Foreign suppliers customize inputs for specific Home producers. These inputs are assumed to be useless to other producers in the benchmark model. Because contracts are incomplete, the parties cannot write an enforceable ex ante contract before the supplier invests.

The timing is:

1. Home producers and Foreign suppliers are randomly matched.
2. The Foreign supplier chooses how much customized input x to produce.
3. The producer and supplier bargain over the input price.
4. The Home producer imports the input and produces the final good.

Interpretation: The model turns offshoring into a problem of relationship-specific investment and bargaining. The central price is not a market-clearing price but a negotiated price inside a bilateral relationship.

5 Free trade and the international hold-up problem

Under free trade, the supplier chooses input production before bargaining. In the Nash bargain, the Home producer receives share α of the surplus and the Foreign supplier receives share $1 - \alpha$.

Because the supplier captures only a fraction of the marginal return from investment, it chooses $\hat{x}$ satisfying

$$(1 - \alpha)y'(\hat{x}) = 1.$$

Since $\alpha > 0$ and $y''(x) < 0$, this implies

$$\hat{x} < x^E.$$

Economic message: Free trade is not efficient because the foreign supplier is held up after making relationship-specific investment. The problem is international because the underinvestment occurs across a border and can be affected by trade taxes or subsidies.

6 Constrained-efficient trade policy

The paper allows governments to use trade taxes or subsidies on both the final good and the intermediate input. Let τ_x^H be the Home import tax on input x , τ_x^F the Foreign export tax on input x , and define the combined input tax as

$$\tau_x \equiv \tau_x^H + \tau_x^F.$$

Let τ_1^H be the Home trade tax on the final good. The joint surplus in the bargaining relationship is

$$J(\tau_1^H, \tau_x, x) = (1 + \tau_1^H)y(x) - \tau_x x.$$

The supplier's investment condition becomes

$$(1 - \alpha)(1 + \tau_1^H)y'(\hat{x}) = 1 + (1 - \alpha)\tau_x.$$

The constrained-efficient policy is:

$$\tau_1^{HE} = 0, \quad \tau_x^E = -\frac{\alpha}{1 - \alpha}.$$

That is, efficient policy keeps free trade in the final good and subsidizes intermediate-input trade.

Interpretation: The input subsidy directly targets the distorted margin. A final-good tariff or subsidy could affect input investment indirectly, but it also distorts consumption. The input subsidy fixes the hold-up problem without creating that final-good distortion.

7 Nash equilibrium trade policy

When governments choose policy noncooperatively, the efficient policy is not obtained.

The paper's Nash result has two key parts:

- Foreign maintains free trade in the final good and taxes exports of the intermediate input.
- Home intervenes in both the input market and the final-good market.

The outcome has two inefficiencies:

1. Intermediate-input trade is inefficiently low: $\hat{x}^N < x^E$.
2. The local final-good price in Home is inefficiently low.

Economic intuition: Governments use trade policy not only to correct the hold-up problem but also to shift surplus across countries. Home distorts the final-good price because doing so affects the bargaining surplus between Home producers and Foreign suppliers. Foreign taxes input exports because it can pass part of the burden onto Home producers through ex post bargaining.

8 Political economy extension

The paper then allows governments to put extra weight on producer surplus. This changes the level and sometimes the sign of policy instruments. For example, sufficiently politically motivated governments may use import tariffs or export subsidies rather than the subsidies/taxes predicted by the benevolent-government version.

However, the main inefficiencies persist:

- Nash policies still produce inefficiently low input trade.
- Nash policies still distort the Home final-good price.

Interpretation: Political economy changes the surface appearance of trade policy but not the deeper institutional problem created by offshoring and bilateral bargaining.

9 Trade agreements: beyond market access

9.1 Why market access is not enough

Traditional GATT/WTO bargaining focuses on market-access commitments. In standard market-clearing models, a market-access commitment can protect the exporting country while leaving the importing country flexibility over its policy mix. This works because policies that leave market access unchanged also leave international prices and trade volumes unchanged.

In the offshoring model, this logic breaks down. Input prices are determined by bargaining, not solely by market clearing. A government can change policies in ways that leave some measure of trade volume unchanged but still alter bargaining surplus, international prices, and welfare.

Result: Efficient trade agreements must achieve **deep integration**: they must directly constrain policies that extend beyond market-access commitments.

9.2 Beyond the terms of trade

In the standard terms-of-trade theory, the basic problem solved by trade agreements is a Prisoners' Dilemma generated by terms-of-trade manipulation. Political economy motives do not change the nature of the problem.

In this paper, political economy motives do change the nature of the problem. A foreign government's distributional goal may be served more efficiently when the Home government adjusts its policies than when the Foreign government relies only on its own export-promotion instruments. This creates a **political externality**: one country's policy mix affects the efficiency with which another country can redistribute income to favored groups.

Thus an efficient trade agreement has two roles:

1. provide escape from a terms-of-trade-driven Prisoners' Dilemma;
2. coordinate policies across countries to reduce deadweight loss from politically motivated export-promotion programs for traded intermediate inputs.

Economic message: Reciprocity and nondiscrimination are no longer enough to guide countries to efficient outcomes. Agreements must be more detailed and more tailored to member-specific needs.

10 Extensions and robustness

The paper studies several extensions. The central message survives each one:

- **Secondary markets:** allowing the input to have some resale value does not eliminate the basic bargaining problem.
- **Ex ante lump-sum transfers:** allowing upfront transfers changes surplus sharing but not the need for deep policy coordination.
- **Vertical integration:** integration changes the internal organization of production but does not remove the trade-agreement problem.
- **Multiple foreign countries and search costs:** the model can be interpreted as a reduced form of costly search for trading partners.
- **Ad valorem tariffs:** the main logic does not depend on using specific taxes.
- **Domestic suppliers and two-sided investments:** introducing additional suppliers or investment margins reinforces the need to consider a wider set of policies.

Interpretation: The mechanism is not an artifact of an extreme benchmark. The important feature is that international prices in offshoring relationships are shaped by lock-in and bargaining.

11 Figures and source panels

The paper contains one conventional figure and no conventional tables. The remaining panels below are direct source extractions from the paper around the key model structure and propositions.

11.1 Figure 1: tariff concessions and input customization

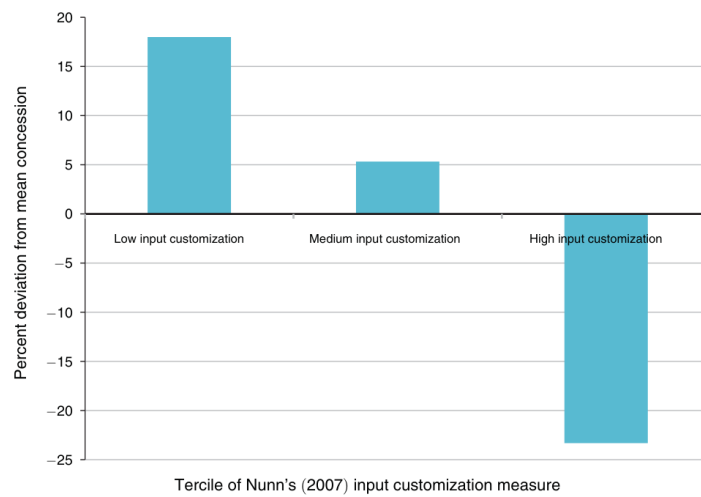


FIGURE 1. PERCENT DEVIATION FROM MEAN CONCESSION BY TERCILE OF INPUT CUSTOMIZATION MEASURE

Figure 1. Percent deviation from mean concession by tercile of input customization measure.

Read: The bars report how WTO tariff concessions differ across sectors with low, medium, and high input customization. Concessions are positive relative to the mean in low-customization sectors, modestly positive in medium-customization sectors, and strongly negative in high-customization sectors.

Economic message: The pattern is consistent with the paper's view that traditional WTO tariff bargaining is harder in sectors where customized inputs and offshoring frictions are more important.

11.2 Model structure and free-trade hold-up

model. We adopt a setting of incomplete contracts between final good producers and input suppliers, where no (enforceable) contracts can be signed between suppliers and producers prior to the initial supplier investment decisions.¹³ Without an initial contract, the price at which each supplier in F sells its inputs to a producer in H is then decided ex post through (Nash) bargaining.

We now describe the structure of the bilateral producer-supplier relationship in detail. We assume that all agents have an ex ante zero outside option. The sequence of events is as follows:

Stage 1: The unit measure of producers in H and suppliers in F are randomly matched, producing a unit measure of matches. Each agent decides whether to stay with his match or exit the market. In the former case, the producer provides the supplier with a list of customized input specifications. In the latter case, each agent obtains his ex ante outside option (equal to zero).

Stage 2: Each supplier decides on the amount x of customized input to be produced (at marginal cost of 1).

Stage 3: Each producer-supplier pair bargains over the price of the intermediate input. We consider the generalized Nash bargaining solution with weights α and $1 - \alpha$ for the home producer and foreign supplier, respectively, where $\alpha \in (0, 1)$.

(recall that the cost of producing x is sunk at this point). Therefore, in the Nash bargain of stage 3, the final good producer in H obtains a payoff equal to $\alpha y(x)$ and the input supplier in F is left with a payoff of $(1 - \alpha)y(x)$.

Next, rolling back to stage 2, observe that the input supplier chooses x to maximize $(1 - \alpha)y(x) - x$, so the optimal quantity $\hat{x}$ of input satisfies

$$(3) \quad (1 - \alpha)y'(\hat{x}) = 1.$$

Given the concavity of $y(x)$, it is clear from a comparison of (3) with (2) that $\hat{x} < x^E$ as long as $\alpha > 0$. This is the under-investment associated with the hold-up problem, and it reflects the fact that the producer and supplier bargain over the price of the input *after* the supplier has already sunk investment in input supply. Only if the supplier were to have full bargaining power ($\alpha \rightarrow 0$) would the hold-up inefficiencies disappear.

Finally, consider stage 1. If the producer hands the supplier a list of customized input specifications, the producer anticipates obtaining a payoff equal to $\pi^H = \alpha y(\hat{x})$, which exceeds the payoff he would obtain by not providing the specifications (recall that the ex ante outside option of producers is equal to 0). Similarly, by agreeing to form a partnership with the home producer, the supplier anticipates obtaining a payoff of $\pi^F = (1 - \alpha)y(\hat{x}) - \hat{x}$, which also exceeds his ex ante outside option.¹⁴ In sum, no separations will occur at stage 1. Note also that the sum of payoffs of the two parties is equal to $y(\hat{x}) - \hat{x}$, which is strictly less than the sum of payoffs that would obtain when investment is chosen at the efficient level x^E defined by (2).

Now consider the measure of social welfare in each country implied by our Benchmark Model. With our assumption of quasilinear preferences, this measure is given by consumer surplus plus profits plus trade tax revenue (the latter being zero under free trade).¹⁵ Using (1), we have that country j 's demand for good 1 is given by $D_j(p_j^1) \equiv u^{j-1}(p_j^1)$, with consumer surplus then defined as $CS^j(p_j^1) \equiv \int_{p_j^1}^{\bar{p}} D_j(p) dp$ where $\bar{p}$ is the "choke" price (if any) for country j 's demand of good 1. World aggregate welfare under free trade may then be represented by

$$\begin{aligned} W^W &= W^H + W^F = CS^H(1) + CS^F(1) + \pi^H + \pi^F \\ &= CS^H(1) + CS^F(1) + y(\hat{x}) - \hat{x}, \end{aligned}$$

which is strictly lower than world welfare in the presence of production efficiency because $y(\hat{x}) - \hat{x} < y(x^E) - x^E$. We summarize this discussion with:

Source panel: bilateral producer-supplier structure and stage game.

Source panel: free-trade investment condition and Proposition 1.

Read: The left panel lays out the incomplete-contracting stage game. The right panel derives the supplier investment condition and states that free trade produces an inefficiently low volume of input trade.

Economic message: The reason free trade fails is not a standard tariff distortion. It is the combination of relationship-specific investment and ex post bargaining.

11.3 Efficient policy and Nash policy

provided that $[y'(\hat{x}) - 1] > 0$ which by (2) implies that $\hat{x} < x^E$: this suggests that an import tariff or export subsidy on trade in the final good 1 could raise welfare in the world, by increasing $\hat{x}$ toward x^E and thereby helping to ameliorate the hold-up problem at the cost of lost consumer surplus. However, recalling from (4) that $\partial \hat{x} / \partial \tau_x < 0$, it is clear from (7) that the optimal choice of τ_x must ensure that $[y'(\hat{x}) - 1] = 0$, thereby achieving productive efficiency: there is no associated loss in consumer surplus when the tariff on imported inputs τ_x is used to increase $\hat{x}$, and the optimal choice of τ_x therefore solves completely the hold-up problem and achieves productive efficiency. This in turn leaves no reason for government intervention with regard to trade in the final good 1. Hence, the optimal choice of τ_1^H is $\tau_1^{HE} = 0$. On the other hand, the constrained-efficient policies do call for intervention with regards to input trade, as long as holdup problems exist, i.e., $\alpha > 0$. In particular, from equation (4) it follows that the optimal trade tax is an input subsidy in an amount equal to $\tau_x^E \equiv \tau_x^{HE} + \tau_x^{FE} = -\alpha/(1 - \alpha)$. We may thus state:

PROPOSITION 2: *In the Benchmark Model, the constrained-efficient trade policy choices maintain free trade in the final good and subsidize importation of the input so as to solve the hold-up problem and achieve an efficient volume of input trade ($\hat{x}^E < x^E$).*

Source panel: constrained-efficient policy and Proposition 2.

level: The reason is that the level of x is chosen by the foreign supplier to maximize foreign profits, and so there is no gain to the foreign country from manipulating this choice with export-sector intervention. Moreover, foreign suppliers do not bear the full burden of the increase in the marginal cost of production associated with an export tax, because they have less than full bargaining power in their negotiations with final good producers ($\alpha > 0$). Hence, F is able to pass part of the cost of the export tax on to the home country while keeping the entire benefit from it (in the form of tax revenue), making an export tax, rather than an export subsidy, the optimal intervention for the foreign government.

We can thus conclude that, when governments choose their policies noncooperatively, international efficiency is not achieved. In particular, we may state:

PROPOSITION 3: *In the Nash equilibrium of the Benchmark Model, F maintains free trade in the final good and taxes the exports of the input, while H intervenes in both the final good and input markets, resulting in (i) an inefficiently low volume of input trade ($\hat{x}^N < x^E$), and (ii) an inefficiently low local price for the final good in H 's market.*

Source panel: Nash policy inefficiencies and Proposition 3.

Read: Proposition 2 says the efficient intervention is free trade in the final good plus an input subsidy that solves hold-up. Proposition 3 says Nash policies generate inefficiently low input trade and an inefficiently low Home final-good price.

Economic message: Trade policy has two roles in this model: it can repair the hold-up problem, but it can also shift surplus across countries. Noncooperative governments overuse the second role.

11.4 Deep integration and political externalities

We summarize this discussion with:

PROPOSITION 4: *In the presence of offshoring, an efficient trade agreement must achieve deep integration, requiring governments to agree to constraints on policies that extend beyond market access commitments.*

Finally, it bears emphasis that, as we indicated earlier, the finding summarized in Proposition 4 applies whether or not governments face political economy pressures.

B. Beyond the Terms of Trade

We next establish that the specific nature of the underlying problem for a trade

by these policy adjustments, an efficiency improvement over the political optimum can be generated.³²

Evidently, then, the home country's policies can help provide a more efficient means of redistributing income toward input suppliers in the foreign country than is possible with the foreign country's own policies alone; and when the foreign government values this redistribution, the need for additional international policy coordination beyond that required to eliminate terms-of-trade manipulation is then implied.³³ We summarize this discussion with:

PROPOSITION 5: *In the presence of offshoring, an efficient trade agreement must serve two roles: it must provide governments with an avenue of escape from a terms-of-trade driven Prisoners' Dilemma; and when the foreign government objectives include political economy considerations, it must coordinate the setting of policies across countries so as to reduce the deadweight loss associated with export promo-*

Source panel: deep integration and Proposition 4.

Source panel: two roles of trade agreements and Proposition 5.

Read: Proposition 4 says efficient trade agreements must constrain policies beyond market access. Proposition 5 says the agreement must also coordinate policies when political economy motives create cross-border redistribution externalities.

Economic message: Offshoring raises the demand for deep integration because the relevant inefficiencies are not exhausted by market-access problems.

11.5 Institutional implication and conclusion

MFN) can help guide countries to politically optimal policies as we have defined those policies above. And it can be confirmed that the properties of reciprocity that accomplish this operate as well in the model that we have developed here.³⁵ But as we have just shown and as Proposition 5 reflects, when governments have political economy motives, politically optimal tariffs are not efficient in the presence of offshoring. And so, in this setting, reciprocity's ability to guide governments to politically optimal policies will not deliver efficient outcomes (except in the case where governments are national income maximizers). Proposition 5 therefore implies that the traditional pillars of the GATT/WTO architecture cannot be counted on to deliver efficient bargaining outcomes in the presence of offshoring. Notice, too, that according to Proposition 5, the problem that a trade agreement must address *varies* with the political preferences of member governments. This carries a further implication: when offshoring is present, simple and general rules that can help governments negotiate to efficient policy choices may simply be unavailable.

Finally, it is worth emphasizing that the statements in Propositions 3, 4, and 5 continue to be true even when the hold-up inefficiencies under free trade in the Benchmark Model disappear. More specifically (and abstracting from political economy forces though analogous statements apply in their presence), recall that when $\alpha \rightarrow 0$, foreign suppliers have full bargaining power and the level of investment under free trade is at its efficient level (i.e., $\hat{x} \rightarrow x^E$), thereby implying that the second-best policies call for no trade intervention ($\tau_1^{HE} = 0, \tau_i^E \rightarrow 0$). Nevertheless, as the expressions in (8) and (13) make clear, $\hat{\tau}_1^{HN} < 0$ and $y'(\hat{x}^N) > 1$ even when $\alpha \rightarrow 0$, and thus Nash policies continue to exhibit the same two inefficiencies in that case. This indicates that the key feature of the Benchmark Model which is responsible

producer's input. The Nash policy choices depart from those in our Benchmark Model, but they do so in an analogous manner to the case with domestic suppliers (see the online Appendix for details).

VI. Conclusion

In this paper, we have initiated the study of trade agreements in the presence of offshoring. Our findings indicate that the rise of offshoring and its implications for international price determination will complicate the task of trade agreements for two reasons. First, the mechanism by which countries shift the costs of intervention on to their trading partners is more complicated in the presence of offshoring and extends to a wider set of policies than is the case when offshoring is not present, and this implies that the agreements themselves must extend beyond market access to a wider set of policies as well—a form of deep integration. And second, the underlying problem that a trade agreement must address in the presence of offshoring varies with the political preferences of member governments, interfering with the ability of governments to rely on reciprocity and non-discrimination to guide their negotiations and shape their agreements.

As a consequence of these findings, we have argued that the growing prevalence of offshoring will make it increasingly difficult for governments to utilize

Source panel: implications for reciprocity, MFN, and WTO architecture.

Source panel: conclusion on deep integration and individualized agreements.

Read: The first panel explains why traditional WTO pillars cannot generally deliver efficient bargaining outcomes in the presence of offshoring. The second panel states the paper's concluding claim that agreements must evolve from market access toward deep integration.

Economic message: The paper's institutional implication is strong: the rise of global value chains pushes the world away from simple, broadly applied rules and toward more policy-specific, member-specific agreements.

12 Short summary

Antràs and Staiger study how offshoring changes trade agreements. In the benchmark model, Home final-good producers buy customized inputs from Foreign suppliers. Because contracts are incomplete, the supplier invests first and bargains later. This creates hold-up and inefficiently low input trade under free trade.

Efficient policy uses an input subsidy to offset the hold-up problem and leaves final-good trade undistorted. But governments acting noncooperatively use policy to shift bargaining surplus, resulting in inefficiently low input trade and a distorted Home final-good price. Political economy motives change the signs and levels of policy instruments but preserve the core inefficiencies.

The paper's main institutional result is that efficient agreements must go beyond market access. In an offshoring world, policies affect welfare through bilateral bargaining, not only through market-clearing terms of trade. As a result, traditional GATT/WTO principles such as reciprocity and nondiscrimination are less reliable guides to efficient agreements.

13 Conclusion

The paper's central insight is that offshoring changes the object of international policy cooperation. When inputs are customized and prices are bargained, trade agreements must deal with relationship-specific investment, surplus division, and domestic policies that affect ex post bargaining. The required agreement is deeper and more tailored than the traditional market-access bargain.

Bottom line: Global value chains make trade agreements more complex because the relevant market failure is not just too little market access. It is too little relationship-specific investment and too much unilateral manipulation of bargaining surplus. This is why offshoring poses a profound institutional challenge for the WTO.